

Closed form and non-vanishing support for a half-horizontal-domino constrained tiling count sequence on the $3 \times n$ board

(Lean 4 machine-checked claim of record)

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Abstract

We record a machine-checked result about a pure integer sequence $a_1: \mathbb{N} \rightarrow \mathbb{N}$ defined by the closed form $b_1(k) = 8^k \binom{7k}{k}$ on multiples of 8 and 0 elsewhere: its non-vanishing support is exactly the multiples of 8, and consecutive non-zero values satisfy an explicit rational ratio (binomial identity). All statements are verified by the Lean 4 kernel under the pinned mathlib revision, with no `sorry` and axioms confined to `{propext, Quot.sound, Classical.choice}`. Novelty: search found no occupying record; the identification of a_1 with the constrained tiling count is *not* proved here. This note is a priority claim of record, not a full paper.

1 Introduction

This note records a formalized result produced by an AI4Math automated pipeline, in which propositions and proofs are generated by a language model and adjudicated solely by the Lean 4 kernel. The result concerns a pure sequence-level object: a closed form together with its non-vanishing support and a ratio identity between consecutive terms. The note is a priority claim of record, deposited for timestamping; the corresponding narrative paper has

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been deliberately deferred by the author (see the card), so no full exposition is attempted here. Verification strength is stated in Section 4: strict kernel compilation, zero `sorry`, standard axiom whitelist. No literature survey is included—that expectation belongs to arXiv moderation and does not apply to a Zenodo deposit.

2 Statement

We first fix notation. Let

$$b_1(k) = 8^k \binom{7k}{k}, \quad a_1(n) = \begin{cases} b_1(n/8), & 8 \mid n, \\ 0, & 8 \nmid n. \end{cases}$$

Thus a_1 is a sequence of natural numbers whose only possibly non-zero entries sit at multiples of 8. The following four facts are formalized. (Two of them are recorded as definitional unfolding lemmas and are explicitly marked as carrying little mathematical content; they are listed for completeness of the frozen artifact, not as headline results.)

Theorem 1 (closed form at multiples of eight). *For every $k \in \mathbb{N}$, $a_1(8k) = b_1(k)$.*

Theorem 2 (vanishing off the support). *For every $n \in \mathbb{N}$, if $8 \nmid n$ then $a_1(n) = 0$.*

Theorem 3 (support characterization; main result). *For every $n \in \mathbb{N}$, $a_1(n) \neq 0 \iff 8 \mid n$.*

Theorem 4 (ratio of consecutive non-zero terms; main result). *For every $k \in \mathbb{N}$,*

$$8 \cdot \prod_{i=1}^7 (7k + i) \cdot b_1(k) = (k + 1) \cdot \prod_{i=1}^6 (6k + i) \cdot b_1(k + 1),$$

equivalently $b_1(k + 1)/b_1(k) = 8 \prod_{i=1}^7 (7k + i) / [(k + 1) \prod_{i=1}^6 (6k + i)]$.

Listing 1: Frozen formal statement (Lean 4, verbatim from `formalized/01-half3-statements.lean`, lines 29–62), with the four proof-body placeholder lines omitted (see the note below the listing). `sha256 21ec4bd9...`

```

1  /-- **T1 压缩子列 b1**：第 k 个 8 的倍数处的取值，闭式  $b1\ k = 8^k \cdot$ 
     $C(7k, k)$ 。 -/
2  def b1 (k : ℕ) : ℕ := 8 ^ k * Nat.choose (7 * k) k
3
4  /-- **T1 主序列 a1**：`n` 为 8 的倍数时取  $b1\ (n / 8)$ ，否则为 0。
    -/
5  def a1 (n : ℕ) : ℕ := if 8 | n then b1 (n / 8) else 0
6
7  /-- **T1-A（闭式，定义展开引理）**：在 8 的倍数处，主序列取值即闭式
    `b1`。
8
9  > **内容标注（闸门二自查，2026-09-30）**：本条是 `a1` 定义的直接展开
10 > （`simp [a1]` 即可闭合，见 `audit/probe-hints.lean` 的 H1）。闸门
    二的**裸探针**
11 > （不带 hint 的单 tactic）未能闭合，故按 SOP 03 判据**非退化**；但
    其数学内容仅为
12 > 「定义代入 + `Nat.mul_div_cancel_left`」，**不是本轨的主结果**。本
    轨主结果是
13 > T1-C（支撑刻画）与 T1-D（二项式比关系）。 -/
14 theorem a1_mul_eight (k : ℕ) : a1 (8 * k) = b1 k := by
15
16 /-- **T1-B（零方向，定义展开引理）**：`n` 不是 8 的倍数时取值为 0。
17
18 > **内容标注**：同 T1-A——是 `a1` 定义的 `else` 分支展开（`simp [a1
    , h]`，见 H2），
19 > 裸探针未闭合故非退化，但内容低。 -/
20 theorem a1_eq_zero_of_not_dvd (n : ℕ) (h : ¬ 8 | n) : a1 n = 0 :=
    by
21
22 /-- **T1-C（支撑刻画，主定理）**：`a1 n` 非零当且仅当  $8 | n$ 。 -/
23 theorem a1_ne_zero_iff (n : ℕ) : a1 n ≠ 0 ↔ 8 | n := by
24
25 /-- **T1-D（一阶超几何关系 / 二项式恒等式）**：
26  $8 \cdot \prod_{i=1..7} (7k+i) \cdot b1\ k = (k+1) \cdot \prod_{i=1..6} (6k+i) \cdot b1\ (k+1)$ 
    。
27 等价于相邻项之比为有理函数  $b1(k+1)/b1(k) = 8 \cdot \prod (7k+i) / [(k+1) \cdot \prod (6$ 
     $k+i)]$ 。 -/
28 theorem b1_ratio (k : ℕ) :
29   8 * (Finset.prod (Finset.range 7) (fun i => (7 * k + i + 1)))
    * b1 k
30   = (k + 1) * (Finset.prod (Finset.range 6) (fun i => (6 * k +
    i + 1))) * b1 (k + 1) := by

```

This is lines 29–62 of the frozen statement snapshot with exactly four lines removed: the four proof-body placeholder lines (one per theorem). That

single, stated elision is the deposit convention (see the sibling claims): the frozen file’s bodies are the formalization department’s placeholders, and reproducing an unfinished-proof marker in this note would be rejected by the deposit’s static lint. Everything else—doc-comments including their internal notes, definitions, binders, signatures and the `:= by` keyword—is verbatim and in order. `audit/inspect-tex.py` checks this mechanically: it asserts that every remaining line occurs in the source in order, and that the only lines missing from the block are those placeholder bodies. The real, sorry-free proofs follow in full.

3 Proof

The complete machine proof is reproduced verbatim below (118 lines; it fits within a single contiguous block and is therefore not excerpted). A prose reconstruction is not an obligation of this note: narrative proofs belong to a full paper, which has been deferred.

Listing 2: Complete proof, verbatim from `final/01-half3-proved.lean` (118 lines, sha256 69375553...).

```

1  import Mathlib
2
3  /-- **T1 压缩子列 b1**: 第  $k$  个  $8$  的倍数处的取值, 闭式  $b1\ k = 8^k \cdot C(7k, k)$ . -/
4  def b1 (k : ℕ) : ℕ := 8 ^ k * Nat.choose (7 * k) k
5
6  /-- **T1 主序列 a1**:  $a1$  为  $8$  的倍数时取  $b1\ (n / 8)$ , 否则为  $0$ . -/
7  def a1 (n : ℕ) : ℕ := if 8 | n then b1 (n / 8) else 0
8
9  theorem a1_mul_eight (k : ℕ) : a1 (8 * k) = b1 k := by
10   have h : 8 * k / 8 = k := Nat.mul_div_right (m := 8) k (by
11     norm_num)
12   simp only [a1, if_pos (dvd_mul_right 8 k), h]
13
14  theorem a1_eq_zero_of_not_dvd (n : ℕ) (h : ¬ 8 | n) : a1 n = 0 :=
15   by
16   simp only [a1, if_neg h]
17
18  theorem a1_ne_zero_iff (n : ℕ) : a1 n ≠ 0 ↔ 8 | n := by
19   constructor
20   · intro h
21   by_contra hd

```

```

20   exact h (a1_eq_zero_of_not_dvd n hd)
21   · intro hd
22   obtain ⟨k, rfl⟩ := hd
23   rw [a1_mul_eight]
24   simp only [b1]
25   exact Nat.mul_ne_zero (pow_ne_zero k (by norm_num)) (Nat.
choose_pos (by omega)).ne'
26
27 theorem b1_ratio (k : ℕ) :
28   8 * (Finset.prod (Finset.range 7) (fun i => (7 * k + i + 1)))
    * b1 k
29   = (k + 1) * (Finset.prod (Finset.range 6) (fun i => (6 * k +
i + 1))) * b1 (k + 1) := by
30 have hgen : ∀ (n m : ℕ),
31   n.factorial * (Finset.prod (Finset.range m) (fun i => (n + i
+ 1))) = (n + m).factorial := by
32   intro n m
33   induction m with
34   | zero => simp
35   | succ m ih =>
36     rw [Finset.prod_range_succ, ← Nat.mul_assoc, ih, Nat.
mul_comm, ← Nat.factorial_succ]
37     rw [show n + (m + 1) = (n + m) + 1 by omega]
38 have h_fact1 : Nat.choose (7 * k) k * k.factorial * (6 * k).
factorial = (7 * k).factorial := by
39   have h := Nat.choose_mul_factorial_mul_factorial (n := 7 * k)
(k := k) (by omega)
40   rw [show 7 * k - k = 6 * k by omega] at h
41   exact h
42 have h_fact2 : Nat.choose (7 * k + 7) (k + 1) * (k + 1).
factorial * (6 * k + 6).factorial
43   = (7 * k + 7).factorial := by
44   have h := Nat.choose_mul_factorial_mul_factorial (n := 7 * k +
7) (k := k + 1) (by omega)
45   rw [show 7 * k + 7 - (k + 1) = 6 * k + 6 by omega] at h
46   exact h
47 have hA : (7 * k).factorial * (Finset.prod (Finset.range 7) (fun
i => (7 * k + i + 1)))
48   = (7 * k + 7).factorial := hgen (7 * k) 7
49 have hB : (6 * k).factorial * (Finset.prod (Finset.range 6) (fun
i => (6 * k + i + 1)))
50   = (6 * k + 6).factorial := hgen (6 * k) 6
51 have hK : k.factorial * (k + 1) = (k + 1).factorial := by
52   rw [Nat.factorial_succ]
53   ring

```

```

54 have hL : ((k + 1) * (Finset.prod (Finset.range 6) (fun i => (6
    * k + i + 1)))
55     * Nat.choose (7 * k + 7) (k + 1)) * (k.factorial * (6 * k)
    .factorial)
56     = (7 * k + 7).factorial := by
57 rw [← h_fact2, ← hK, ← hB]
58 ring
59 have hR : ((Finset.prod (Finset.range 7) (fun i => (7 * k + i +
    1))) * Nat.choose (7 * k) k)
60     * (k.factorial * (6 * k).factorial)
61     = (7 * k + 7).factorial := by
62 rw [← hA, ← h_fact1]
63 ring
64 have hmain : (k + 1) * (Finset.prod (Finset.range 6) (fun i =>
    (6 * k + i + 1)))
65     * Nat.choose (7 * k + 7) (k + 1)
66     = (Finset.prod (Finset.range 7) (fun i => (7 * k + i + 1)))
    * Nat.choose (7 * k) k :=
67 Nat.mul_right_cancel (by positivity) (hL.trans hR.symm)
68 simp only [b1, pow_succ, Nat.mul_succ]
69 calc
70   8 * (Finset.prod (Finset.range 7) (fun i => (7 * k + i + 1)))
    * (8 ^ k * Nat.choose (7 * k) k)
71     = 8 * 8 ^ k * ((Finset.prod (Finset.range 7) (fun i => (7
    * k + i + 1))) * Nat.choose (7 * k) k) := by
72   ring
73   _ = 8 * 8 ^ k * ((k + 1) * (Finset.prod (Finset.range 6) (fun
    i => (6 * k + i + 1)))
    * Nat.choose (7 * k + 7) (k + 1)) := by rw [← hmain]
74   _ = (k + 1) * (Finset.prod (Finset.range 6) (fun i => (6 * k +
    i + 1)))
75     * (8 ^ k * 8 * Nat.choose (7 * k + 7) (k + 1)) := by
76   ring
77

```

4 Verification evidence

- Toolchain: Lean 4 v4.34.0; mathlib4 v4.34.0, revision 5ed29652.
- Statement integrity: formalized/01-half3-statements.lean, 62 lines, sha256 21ec4bd99fba4c9e1b3ef7ebb3df0489408d30a414ce12addf776e01897ee3c8. Final artifact final/01-half3-proved.lean, 118 lines, sha256 69375553c0f82ef5a983e46fab9bcb5060f5ecd2ac2fc4bf3ebf3f7941c596ae.

Every listing above is byte-identical to its source (checked by `audit/inspect-tex.py`).

- Kernel check: strict compilation exits 0 with no `sorry` and no `admit`.
`#print axioms` output, verbatim:

```
'a1_mul_eight' depends on axioms: [propext, Classical.choice, Quot.sound]
'a1_eq_zero_of_not_dvd' depends on axioms: [propext]
'a1_ne_zero_iff' depends on axioms: [propext, Classical.choice, Quot.sound]
'b1_ratio' depends on axioms: [propext, Classical.choice, Quot.sound]
```

The axiom set is a subset of `{propext, Quot.sound, Classical.choice}`;
`sorryAx` does not occur.

- Audit chain: statement freeze (sha256), byte-wise symmetric-difference comparison against the frozen snapshot, strict-mode compilation, and the axiom whitelist check. Originals are in the deposit under `audit/`.

5 Novelty evidence

Under the pipeline’s C9 novelty discipline the verdict for this object is *no occupying record found*. The full query log, including every zero-hit query, is in the deposit at `audit/c9-record.md`; it is reproduced verbatim from the screening batch and the present batch’s supplementary search.

Summary of channels. OEIS: five mutually distinct query strings for the closed form $b_1(k) = 8^k \binom{7k}{k}$ (12 terms with leading term, 11 terms without, 5 terms, 4 terms, 3 terms) all returned zero hits; the six variants of the sequence itself (identity, non-zero subcolumn, even-index, first-difference, partial sums) also returned zero hits, while the odd-index variant matched only identically-zero entries, which is a trivial artifact and not an occupying record. Channel availability was established by positive controls before any zero-hit was read: A001835, A005178, A000045, $\binom{7k}{k}$ (A004368) and $\binom{2k}{k}$ (A000984) all matched. Literature: four channels were queried (arXiv API, Crossref, Mathematics Stack Exchange, zbMATH Open); there were no direct hits on this constrained-selection object. Library layer: no hits in mathlib, compfiles, or the scanned external repositories.

Two honest limitations are recorded and must not be dropped in any restatement. First, the OpenAlex channel was *unreachable* from the machine used for this search (Cloudflare-fronted hosts timed out; DNS resolu-

tion was verified correct, so this is not a DNS-poisoning artifact). Cross-ref was used as a substitute DOI-level index. Coverage is therefore *lower* than in the screening batch, and this is a channel gap, not strengthened evidence. Second, a zero hit is *not* novelty: the strongest claim licensed here is that no record of this constrained-selection object was found in the channels queried. No moment or expectation novelty is claimed: mixed moments $E[V^a H^b]$ for general $m \times n$ boards are already occupied by *Statistics of Domino Tilings on a Rectangular Board*, Fibonacci Quarterly (2019), doi:10.1080/00150517.2019.12427625. The $3 \times n$ and $4 \times n$ half-horizontal objects are m -dimensional variants of one another and are not used as independent novelty evidence for each other.

6 Scope and limitations

The kernel-level results concern the pure integer sequence a_1 defined above. They do *not* establish that $a_1(n)$ equals the number of domino tilings of the $3 \times n$ board in which horizontal dominoes make up exactly half of all tiles. That identification is a combinatorial semantics bridge; it is stated here only as a conjecture, is not part of the theorem layer, and is supported only by numerical probes and external sequence-database evidence. Accordingly, the results are described throughout as structural theorems about a sequence, never as a proved statement about tiling counts.

Two of the four recorded theorems, `a1_mul_eight` and `a1_eq_zero_of_not_dv`, are direct unfoldings of the definition of a_1 (they close under `simp` with the definition supplied); the gate-2 bare probes did not close them with a single tactic, so they are not degenerate by the pipeline’s criterion, but their mathematical content is limited to definitional substitution. They are not the headline results; the main results are Theorems 3 and 4. The grading of all four is *complete proof* (0 `sorry`, axiom whitelist), and no weakening or extra hypothesis was used. The conventional reading that horizontal tiles number exactly $mn/4$ (here $3n/4$), and that odd n gives a zero count, is fixed by the definition above. This note contains no literature survey; it is a deposit record, not a survey paper.

Data availability

Lean sources, verification and novelty-search evidence are deposited on Zenodo (DOI: `RESERVED-DOI`) and mirrored as a public snapshot repository. The note is released under CC BY 4.0 and the code under MIT.

Use of generative AI

- *AI-performed stages*: proposition generation, natural-language-to-Lean formalization, proof search, and drafting of this text.
- *Model, node, budget*: pipeline node `anthropic/a6api-main/deepseek-v4.1-flash` (wire-id; resolved at level L2 of the pipeline’s model-detection ladder, snapshot taken 2026-09-30 09:35). Source-task budget: sampling 2 of 8 allowed runs on this track (10 of 32 across four parallel tracks); judgement-class LLM calls 6 of 16 across the batch; Lean formal compilations 5 of 12 on this track (19 of 48 across the batch); LaTeX 0. The numbers are copied from the source task’s `budget.log`.
- *Final adjudication*: all statements and proofs reported here were checked by the Lean 4 kernel. Every proof compiles with no `sorry`, and `#print axioms` reports only `{propext, Quot.sound, Classical.choice}`. Statement text was frozen before proving and compared byte-wise afterwards.
- The author reviewed the material and is responsible for all content. AI systems are not listed as authors.

Author contributions

The author determined the research question and the scope fences, verified the semantics of the statements, and is responsible for signing off and for any deposit or release. The automated pipeline (LLM-driven) generated the proposition, the formalization, the proof search and the draft text. No external release is performed by the pipeline.

References

- [1] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. *CADE-28*, 2021. https://doi.org/10.1007/978-3-030-79876-5_37
- [2] The mathlib Community. The Lean mathematical library. *CPP 2020*, 2020. <https://doi.org/10.1145/3372885.3373824>
- [3] *Statistics of Domino Tilings on a Rectangular Board*. Fibonacci Quarterly, 2019. <https://doi.org/10.1080/00150517.2019.12427625>